

Lorentz invariance

Let X be a space-like vector. There exists a Lorentz transformation Λ such that the time component vanishes in the transformed vector $X' = \Lambda X$.

For example, let X be the space-like vector

$$X = \begin{pmatrix} 2 \\ 0 \\ 0 \\ 3 \end{pmatrix}$$

Verify X is space-like.

$$X^\mu X_\mu = 2^2 - 3^2 = -5$$

We seek Λ such that

$$X' = \Lambda X = \begin{pmatrix} \gamma & 0 & 0 & -\beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\beta\gamma & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} 2 \\ 0 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ z' \end{pmatrix}$$

where $0 \leq \beta < 1$ and

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}}$$

This reduces to finding β such that

$$2\gamma - 3\beta\gamma = 0$$

Solve for β .

$$\beta = \frac{2}{3}$$

Hence

$$\Lambda = \begin{pmatrix} 3/\sqrt{5} & 0 & 0 & -2/\sqrt{5} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -2/\sqrt{5} & 0 & 0 & 3/\sqrt{5} \end{pmatrix}$$

and

$$X' = \Lambda X = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \sqrt{5} \end{pmatrix}$$

Eigenmath script